

The Black-Scholes Model (2. Transition Density)

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

The one-dimensional heat equation

Consider a rod where the temperature at any location x at any time t is given by the function $u(x, t)$.

Let's assume the initial temperature profile of the rod is given as $u(x, 0) = f(x)$. The **heat equation** or the **diffusion equation** that describes how the temperature evolves at different points on the rod is given by the equation below

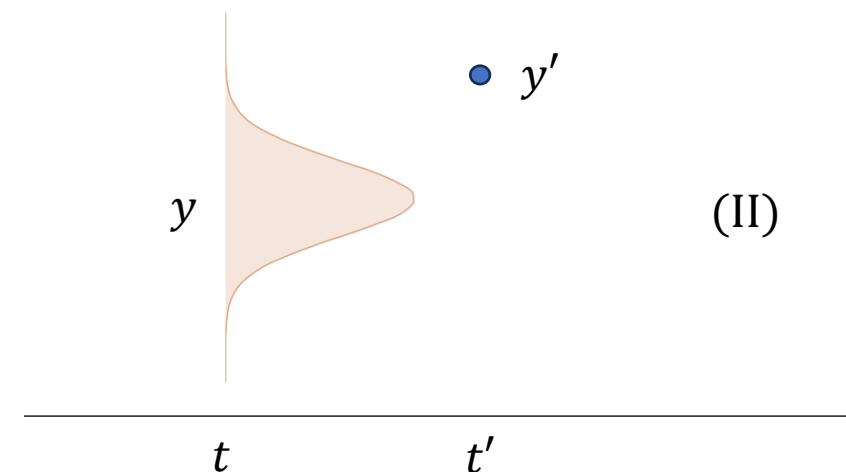
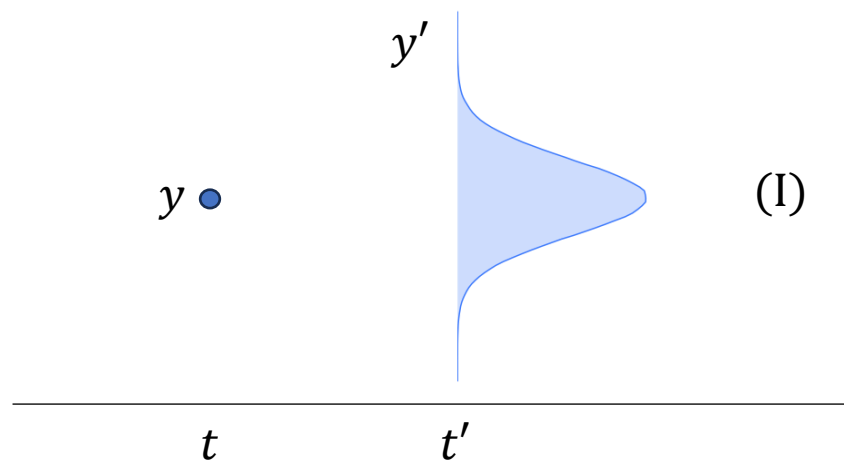
$$\frac{\partial u}{\partial t} = \kappa \left(\frac{\partial^2 u}{\partial x^2} \right)$$

κ is the property of the material. This equation is a linear parabolic type of PDE and is of special importance in Quant-finance. The celebrated Black-Scholes PDE is reducible to the heat equation.

The transition density

Consider a stochastic process Y_t . Let's say the value of the process at t is y and the value at $t' > t$ is y' . We want to know two things

- ▶ I. Given (y, t) what is the probability of (y', t')
- ▶ II. Given (y', t') what is the probability of (y, t)



The transition density

Let's denote the **transition density function** as $p(y, t ; y', t')$. The semi-colon separates the group of variables based on state and time. This quantity satisfies two equations

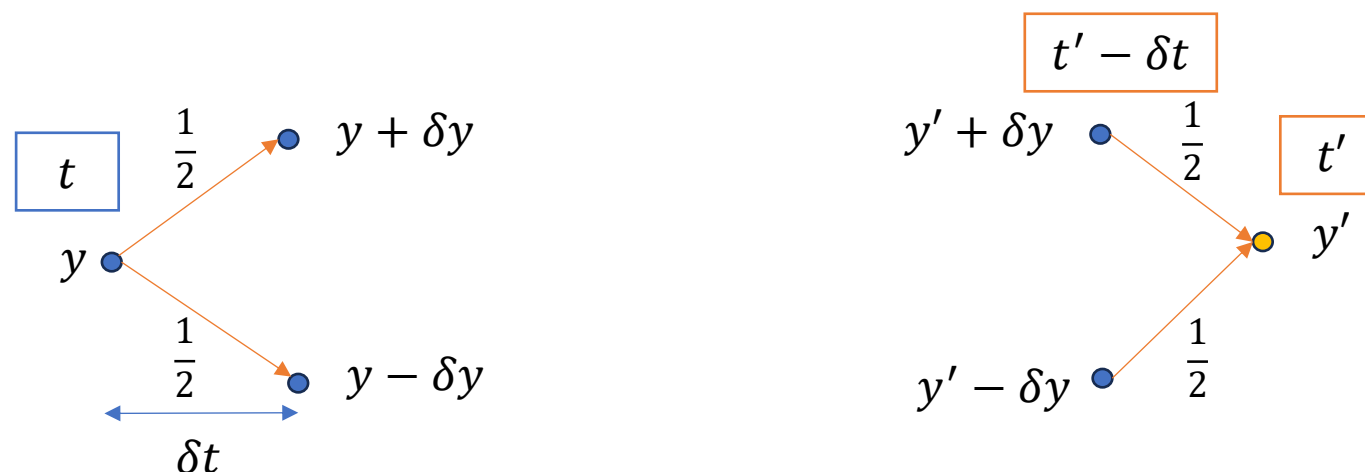
- ▶ I. **Kolmogorov Forward equation** (keep y, t fixed, vary y', t')
- ▶ II. **Kolmogorov Backward equation** (keep y', t' fixed, vary y, t)

The forward equation is also called as **Fokker-Planck** equation.

Let's derive the equations for a simple random walk process.

The Simple Random Walk

Consider a simple random walk process. The value at time t is y . At each time step δt , the increment can be either $+\delta y$ or $-\delta y$



The probability of reaching (y', t') from (y, t) can be written in recursive form as
 probability of first reaching $(y' + \delta y, t' - \delta t)$ and then (y', t') + probability of first reaching $(y' - \delta y, t' - \delta t)$ and then (y', t')

$$p(y, t; y', t') = \frac{1}{2}p(y, t; y' + \delta y, t' - \delta t) + \frac{1}{2}p(y, t; y' - \delta y, t' - \delta t)$$

The Simple Random Walk

Let's write down the Taylor Series expansion of the following

$$p(y, t ; y' + \delta y, t' - \delta t) = p(y, t ; y', t') + \frac{\partial p}{\partial y'} (\delta y) + \frac{\partial p}{\partial t'} (-\delta t) + \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'} (\delta y)^2 + \dots$$

$$p(y, t ; y' - \delta y, t' - \delta t) = p(y, t ; y', t') + \frac{\partial p}{\partial y'} (-\delta y) + \frac{\partial p}{\partial t'} (-\delta t) + \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'} (\delta y)^2 + \dots$$

Substituting these in the equation

$$p(y, t ; y', t') = \frac{1}{2} \left[p(y, t ; y', t') + \frac{\partial p}{\partial y'} (\delta y) + \frac{\partial p}{\partial t'} (-\delta t) + \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'} (\delta y)^2 \right] +$$

$$\frac{1}{2} \left[p(y, t ; y', t') + \frac{\partial p}{\partial y'} (-\delta y) + \frac{\partial p}{\partial t'} (-\delta t) + \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'} (\delta y)^2 \right]$$

We get

$$\frac{\partial p}{\partial t'} \delta t = \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'} (\delta y)^2$$

The Fokker-Planck Equation

As $\delta t \rightarrow 0$, the ratio of $\frac{(\delta y)^2}{\delta t}$ should be finite for the equation to make sense. Let's call this ratio as

$$\frac{1}{2} \frac{(\delta y)^2}{\delta t} = c^2$$

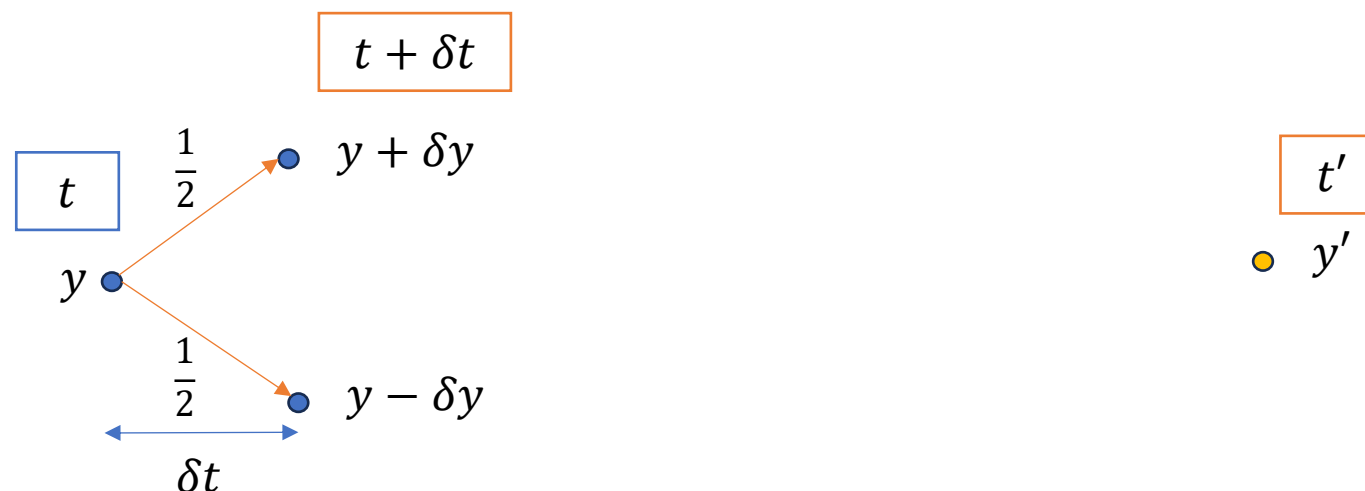
We get

$$\frac{\partial p}{\partial t'} = c^2 \frac{\partial^2 p}{\partial^2 y'}$$

This is the **Kolmogorov-Forward Equation** or **Fokker-Planck equation** which describes the evolution of transition density at any future time.

The Backward Equation

We can build the recursive equation another way



The probability of reaching (y', t') from (y, t) can be written as

probability of first reaching $(y + \delta y, t + \delta t)$ and then (y', t') + probability of first reaching $(y - \delta y, t + \delta t)$ and then (y', t')

$$p(y, t ; y', t') = \frac{1}{2} p(y + \delta y, t + \delta t ; y', t') + \frac{1}{2} p(y - \delta y, t + \delta t ; y', t')$$

The Backward Equation

Following in similar manner, we find the backward equation is

$$\frac{\partial p}{\partial t} + c^2 \frac{\partial^2 p}{\partial^2 y} = 0$$

Note that the difference is just a change in sign. It's intuitive because the density function depends on the time allowed. Increasing t' has the same effect as decreasing t

As a next step, let's try to find a solution the forward equation.

Solving the Forward Equation

Consider a solution of the form

$$p = t'^a f\left(\frac{y'}{t'^b}\right) \quad x = \frac{y'}{t'^b}$$

Take partial derivative w.r.t y'

$$\begin{aligned} \frac{\partial p}{\partial y'} &= t'^a \times \frac{df}{dx} \times \frac{\partial x}{\partial y'} = t'^a \times \frac{df}{dx} \times \frac{1}{t'^b} \\ &= t'^{(a-b)} \times \frac{df}{dx} \end{aligned}$$

Take 2nd partial

$$\frac{\partial^2 p}{\partial y'^2} = t'^{(a-2b)} \times \frac{d^2 f}{dx^2}$$

Solving the Forward Equation

Now take partial derivative w.r.t t'

$$\begin{aligned}\frac{\partial p}{\partial t'} &= at'^{a-1} \times f(x) + t'^a \times \frac{df}{dx} \times (-y'bt'^{-b-1}) \\ &= at'^{a-1}f(x) - by't'^{a-b-1} \times \frac{df}{dx}\end{aligned}$$

Substituting these in the forward equation

$$\begin{aligned}at'^{a-1}f(x) - by't'^{a-b-1} \frac{df}{dx} &= c^2 t'^{(a-2b)} \times \frac{d^2 f}{dx^2} \\ af(x) - bx \frac{df}{dx} &= c^2 t'^{(1-2b)} \frac{d^2 f}{dx^2}\end{aligned}$$

Solving the Forward Equation

If we look at the last equation the left-hand side is purely a function of x , so the right-hand side should not have anything other than x

$$af(x) - bx \frac{df}{dx} = c^2 t'^{(1-2b)} \frac{d^2 f}{dx^2}$$

This leads to $b = \frac{1}{2}$

$$af(x) - bx \frac{df}{dx} = c^2 \frac{d^2 f}{dx^2}$$

Also note that $\int_{-\infty}^{\infty} p dy' = 1 \Rightarrow \int_{-\infty}^{\infty} t'^a f\left(\frac{y'}{\sqrt{t'}}\right) dy' \quad \text{Substitute} \quad y' = u\sqrt{t'} \quad dy' = \sqrt{t'} du$

$$= t'^{a+\frac{1}{2}} \int_{-\infty}^{\infty} f(u) du$$

Solving the Forward Equation

Note that the expression below is a PDF at all times t'

$$= t'^{a+\frac{1}{2}} \int_{-\infty}^{\infty} f(u) du$$

It's only possible if $a = -\frac{1}{2}$

This leads to
$$-\frac{1}{2}f(x) - \frac{1}{2}x \frac{df}{dx} = c^2 \frac{d^2f}{dx^2} \Rightarrow -\frac{1}{2} \frac{d(xf(x))}{dx} = c^2 \frac{d^2f}{dx^2} \Rightarrow -\frac{1}{2}xf(x) = c^2 \frac{df}{dx}$$

$$\Rightarrow c^2 \frac{d(\ln f)}{dx} = -\frac{1}{2}x$$

$$\Rightarrow \ln f(x) = -\frac{x^2}{4c^2} + \text{Const.}$$

$$\Rightarrow f(x) = A \exp\left(-\frac{x^2}{4c^2}\right)$$

Solving the Forward Equation

Since

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} A \exp\left(-\frac{x^2}{4c^2}\right) dx = 1$$

$$A = \frac{1}{2c\sqrt{\pi}}$$

Using the result

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Therefore

$$f(x) = \frac{1}{2c\sqrt{\pi}} \exp\left(-\frac{x^2}{4c^2}\right)$$



Solving the Forward Equation

Substituting x back

$$p = \frac{1}{2c\sqrt{\pi t'}} \exp\left(-\frac{y'^2}{4c^2 t'}\right)$$

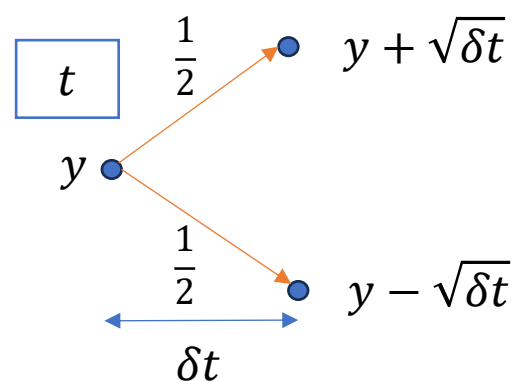
Recognize this ! Substitute , $\sigma = c\sqrt{2t'}$

$$p = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y'^2}{2\sigma^2}\right)$$

The transition density is normal with variance that scales linearly with time. For Brownian Motion, we know $\sigma = \sqrt{t'}$, so $c = 1/\sqrt{2}$

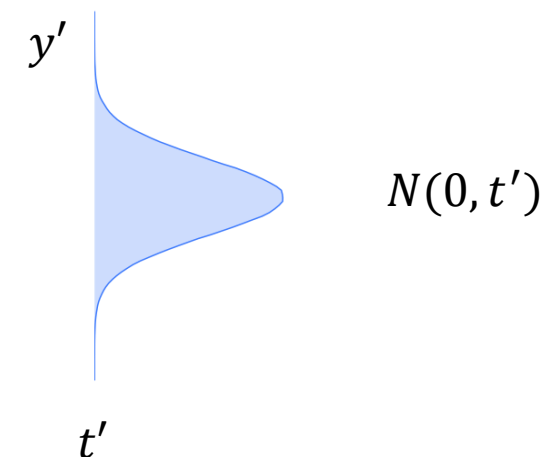
Transition density of Brownian Motion

We have seen that the scaled symmetric random walk approaches Brownian Motion in limit. Let's reconcile the behavior with the forward equation we developed.



$$\frac{1}{2} \frac{(\delta y)^2}{\delta t} = c^2 = \frac{1}{2}$$

(scaled symmetric random walk)



Forward equation for Brownian Motion is $\frac{\partial p}{\partial t'} = \frac{1}{2} \frac{\partial^2 p}{\partial^2 y'}$

Transition density (general)



If we assume that at state is y at t , we can generalize the transition density to reflect the initial condition

$$p(y, t; y', t') = \frac{1}{2c\sqrt{\pi(t' - t)}} \exp\left(-\frac{(y' - y)^2}{4c^2(t' - t)}\right)$$

Summary



- ▶ We started with a simple random walk process and arrived at a very simple PDE similar to the heat equation.
- ▶ We want to enhance the PDE infrastructure for more general processes such as arithmetic Brownian, geometric Brownian, mean reverting process etc....